Blind Assistance

Final Year Project

Session 2018-2022

A 4th Year Student

A project submitted in partial fulfilment of the
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of
BSc. (Hons.)BS in Computer Science / Software Engineering
(CUI)

Department of Computer Science

COMSATS University Islamabad, Lahore Campus

29 Dec 2021

Project Detail

Type (Nature of project)	[✓] **D**evelopment [ ] **R**esearch [ ] **R&D**			
Area of specialization				
Project Group Members				
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Plagiarism Free Certificate

This is to certify that, I am ___Raja Ali Hassan_____ S/D/o ___Karamat Hussain_____, group leader of FYP under registration no CIIT/SP18-BCS-020_/LHR at Computer Science Department, COMSATS University Islamabad, Lahore Campus. I declare that my FYP proposal is checked by my supervisor and the similarity index is _____% that is less than 20%, an acceptable limit by HEC. Report is attached herewith as Appendix A.

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Name of Supervisor: ___Amaid Zia_____ Co-Supervisor (if any):_____

Designation: _____Lecturer_____ Designation: _____

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Abstract

For the final year project, we are developing a navigation mobile application for Android. This app shall be specifically useful for blind people to freely move around in campus independently and without the help of any other person. The application shall use voice and gestures as input to give commands and give outputs through voice as well as Google AR core API.

This application shall ensure that every human being is provided with an equal opportunity for Education despite their physical disabilities.

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1 Introduction

1.1 Introduction

The Blind Assistance application is for Android only. Blind Assistance targets the 'Special people' as an audience, Blind or visually impaired people to be specific. Blind Assistance shall help the visually impaired people to either move from one desired point in the campus to another and navigate towards their class at the specific time.

Blind Assistance application mainly communicates back and forth with the user through voice; however, some inputs are given using gestures as well.

As the Blind Assistance application is opened, the user shall be prompted to enter sign up details. This shall only be done once and saved for future use until the user wishes to use a different account altogether. Now the user would be at the 'Home Page'. Here the user shall have a choice to either go to a desired location or follow the timetable, he/she would have to draw the relevant gesture for desired option. Whatever option is chosen by the user, the user shall be taken to the desired location through 'speech' guidance using Google Maps and/or AR core API using anchors.

1.2 Goals and Objectives

Following are the goals and objectives for our Blind Assistance Final Year Project.

- Benefit special people communities.
- Help each partially or completely blind person to be able to move through campus like any other person, without any kind of outside help.
- Follow classes according to the timetable.
- Go to parking from working space and vice versa.
- Enable special people to be able to earn and support their families by getting equal education without any hindrances.
- Enable blind person to move anywhere as, Blind Assistance shall be implemented for specific locations at Comsat University Islamabad, Lahore Campus only for now but it can be used and extended for any public place, office and institution etc. to support the special people earn their livelihood without the fear of their disability.
- Develop such an application that no one is left behind.

1.3 Problem Statement

A major problem faced by blind people was that their disability was holding them back. They were not able to utilize their full potential to get education, contribute and serve the society. Due to their fear of disability and or being unable to move freely, they would either hire another person to help them move and navigate around or completely avoid attending institutions. In order to counter this and allow special people to be able to overcome this problem, the Blind Assistance application is being developed.

1.4 Assumptions and constraints

Assumptions

- Blind Assistance application shall be available for free.

- Blind Assistance shall guide users through Google maps and AR core API throughout campus.
- Blind Assistance shall make blind person completely free for any fear of disability.

- Blind person shall be able to move at N block and parking area on campus, as per his/her desire without the need for any outside help.
- Blind person shall be guided directly and exactly from class to class as per the user's specific timetable.

Constraints:

- Blind Assistance shall only be available for Android only.
- Only blind people shall be able to benefit from this application.
- This app shall only cover specific areas of Comsat university Islamabad, Lahore campus.
- This application shall only navigate the user through pre-set locations.

1.5 Project Scope

The main purpose that this Final Year Project serves is to ensure that every person can get quality education without any problems. As there were not many apps designed locally to help blind people to move around freely, this project shall cover this void and provide an app specifically for blind people.

The Blind Assistance application aims to make blind people completely independent. By using their voice and a few easy to learn gestures, they would be able to navigate around freely.

2 Requirement Analysis

2.1 Existing System Study

The Blind Assistance application that shall only be available for android only shall make a blind person completely independent of another person's help and his/her fear of the disability. As the app will be completely voice activated, with few inputs given as gestures, the user, upon his own wish, shall be able to give command to the application to either follow his/her timetable or go to a desired location on the campus (In future work).

The app shall use Google maps APIs and AR core anchors to navigate around the campus and guide users through voice activation. The user can also pause the navigation wherever he/she wants to and change the destination location whenever he/she wants to or simply cancel the current route.

The user can go to places like cafes, washrooms, student service centres, different departments and blocks by first making a gesture and then simply speaking into the app when prompted.

2.1.1 Ariadne-GPS

Available for both IOS and Android users, Ariadne-GPS is a somewhat similar app Blind Assistance and helps the visually impaired people to move freely. The interface for Ariadne-GPS is user-friendly hence easy to understand and use. Ariadne-GPS is not for some specific place or area, it can be used anywhere and utilizes Google Maps. [1]

2.1.2 Blind-Square

Blind-Square helps user navigate by speaking to them about their locations and destinations and directions towards them. This application also informs the user about its surroundings. [2]

2.1.3 Moovit

This application is targeted to enable blind people use public transport. Moovit gives detailed analysis of different routes and vehicles available for transportation and their specific timings for different routes. It also provides information about traffic on different routes. [3]

2.1.4 Microsoft Soundscape

Now, IOs users can only access Microsoft Soundscape. Microsoft Soundscape uses a bit different approach. A 3D audio technology is used in this app to identify different aspects of the environment and surroundings of the user, so he/she can familiarize itself with it. This helps the user to find bearings to the identified intersections. [4]

2.1.5 Evelity

Users can move through and navigate using this app through different types of complexes whether big or small such as libraries, malls universities etc. Voice over and talk back features are also part of this app.

2.2 Stakeholder's list

- Developers

- Project Supervisor
- Final Year Project Committee
- Final Year Project Jury
- CUI
- User

2.3 Requirement Elicitation

2.3.1 Functional Requirements

Functional Requirements Table

Requirement No.	Functional Requirements
FR-1.1	When user first time opens the application, User should be guided through the process of registration.
FR-1.2	When user opens app after one-time-registration sequence of gestures should be demanded in order to provide access.

FR-1.3	User should be given a tutorial for understanding of the application usage
FR-2.1	For the Custom destination selection made by user, user is asked whether he/she is inside N block or outside, if outside, user is guided till N block through Google Maps, then inside N block user is guided through AR core.

FR-1

FR-2.3	When user arrives at the destination , the text to speech prompts the user that the destination has arrived, and opens the home page.
FR-3.1	For following timetable option, the app asks if the user is inside or outside N block, if inside, it guides user through AR core towards the next class. If outside, the user is first guided to N block using Google Maps.
FR-3.2	After the app informs user about custom destination, the app asks if the user is inside or outside N block, if inside, it guides user through AR core towards the next class. If outside, the user is first guided to N block using Google Maps.
FR-3.3	When user arrives at the destination, the text to speech prompts the user that the destination has arrived, and opens the home page.
FR-4.1	If user selects the Edit Timetable option, app asks user to speak day of class, after user speaks the day, the app asks for time of slot, after user speaks time, the app repeats the time and day of slot so user can make sure he is in right option of slot.

2.3.2 Non-Functional Requirements

Non functional Requirements 2

NFR-1.1	The Login page of the application will not take more than 3 seconds to load

NFR-1.2	The main screen will take 2 seconds for loading
NFR-1.3	User's Login request with sequence of gestures should take less than 3 sec for verification
NFR-1.4	User should be able easily open the application through Google Voice Assistant preinstalled in the phone.
NFR-2.1	Average load time searching by location will not take more than 10 seconds to load
NFR-2.2	New location added by the user should be added with 3 sec after selection
NFR-2.3	User should be able to hear voice through Phone Speaker and Handsfree/Headphone
NFR-2.4	In case of an inaccurate gesture user should be notified properly
NFR-3.1	Classes location as per the timetable should be kept UpToDate in the database
NFR-4.1	Before updating of a particular class, it should be verified by user
NFR-4.2	Before over writing a particular slot, it should be verified by the user
NFR-4.3	An anchor should not take more than 5 secs to load in session.

2.4 Requirement Traceability Matrix

Requirement traceability matrix 3

Requirements Traceability Matrix	
Project Name	Blind Assistance Application
Project Description	The Project Aims to provide the visual assistance to blind students all around Comsats University Lahore Campus. Depending on the destination entered by user the app will guide user from current location to class, cafeteria, teacher's office or Student Services centre. For the first time when the app is opened the app requires sequence of gestures and logs in.

	app will ask user through voice to enter a gesture to select the navigation type that can be 'Follow the timetable' or 'Enter Custom Location'. Once user selects the Particular navigation type, app inquires user for current location (inside N block, outside N block). When user reaches the destination, the app announces that you have arrived at destination. While guiding user app also informs user about staircases in way. For the following timetable option, user have to save the timetable with the option of edit timetable, in the page app asks for time slot and day he wants to edit, user will add the subject details and Class Name in Page, in this way the timetable is stored in the app.

Requirement ID	Requirements Description	WBS Deliverables	Activity Diagram	Sequence Diagram	Use Case	Test case
FR-1.1	Allows user to register in the Application	1	AC-1.1	SQ-1.1	UC-004	TC-02
FR-1.2	Allows user sign in using gestures sequence	1	AC-1.2	SQ-1.2	UC-001	TC-03
FR-3.1	Allows user to enter and follow the custom location	1.3	AC-3.1	SQ-3.1	UC-011	TC-07
FR-4.1	Allows user to follow location for upcoming	1.3	AC-4.1	SQ-4.1	UC-005	TC-06
	class on timetable					

2.5 Use case descriptions

Use case table 4

ID 001 Use case Name: Login Screen
Priority High

Actors: User	
Use Case Summary	The user will enter the sequence of gestures
Pre-condition:	First need to install the application and ArCore then User will have to login.

Normal Flow of Events	**Alternative Path**
1. The use case will start when the user Opens the application	
2. User will login in by entering a sequence of gestures	

3. User will have to put 3 gestures in a particular order set at the time of sign-up/login.	
4. The sequence will be validated with the provided sequence from the database	If user enters incorrect sequence the login will fail and system will speak the message of invalid input, and prompt to re-enter gestures.

6. The User will be directed to the homepage	.
Post Conditions	
Step#	**Description**
1.	The User will be redirecting their respective Home Page
Use Case Cross References	
Includes	1- Verify Login Details 2- Gesture Invalid sequence Error
Extends	

UC 2- 5

Use case ID 003 Use case Name: Verify Login Details
Priority High
Actors:

Use Case Summary	The user's entered sequence of gesture is compared with the stored sequence from the database
Pre-condition:	First need to provide 3 gestures in a particular order

Normal Flow of Events	**Alternative Path**
1. The use case will start when has provided 3 gestures for login	
2. Sequence and gestures will be compared with the stored ones	
3. The User will be directed to the homepage if it is correct	User will be prompted to re-enter the gestures in correct order.
Post Conditions	
Step#	**Description**
1.	The User will be redirecting their respective Home Page

Use Case Cross References	
Includes	None
Extends	None

UC 3- 6

Use case ID 004 Use case Name: Sign - Up	
Priority: High	
Actors: User	
Use Case Summary	When user opens the app for the first time enters sign up gesture, then enters login details.
Pre-condition:	Users need to open app and he/she must not be signed in.
Normal Flow of Events	**Alternative Path**
1. The use case starts once app is opened, selects the Sign-Up Option by giving relevant gesture.	

2. User need to enter roll number through speech.	
3. When user reaches the password field the new blank page appears, here user can add a sequence of gestures as a password	

UC 4- 7

Post Conditions	
Step#	**Description**
1.	Add new Data base action will be triggered and user details will be saved in database and home page will open
Use Case Cross References	
Includes	N/A
Extends	N/A

UC 5- 8

Use case ID 005 Use case Name: Follow timetable

Priority High	
Actors: User	
Use Case Summary	The user's entered gesture to start following the timetable. User will be directed to the next scheduled slot
Pre-condition:	Input the chosen gesture
Normal Flow of Events	**Alternative Path**
1. The use case will start when input the Timetable following gesture	
2. User will be directed towards the next class	
Post Conditions	
Step#	**Description**

1.	User will start the navigation from the current location to the next class
Use Case Cross References	
Includes	Fetch timetable from Database Inquire about current location
Extends	None

UC 6- 9

Use case ID 006 Use case Name: API UPDATE	
Priority: High	
Actors: Admin	
Use Case Summary	The admin will keep track of latest and suitable APIs for the system and will update them and will publish the updates on Play Store so users can have latest updates, and this can improve their location accuracy, gesture, voice commands and google assistance.

Pre-condition:	Admin must have known how of updating APIs
Normal Flow of Events	**Alternative Path**
1. The app will improve experience of Gesture input, google assistant, voice commands and location preciseness, and directions will get better	The wrong API or previous version if added, this can affect the preciseness and accuracy of app

Post Conditions	
Step#	**Description**
1.	Updated version and its details are published on Play Store so user can update app. Once he/she updates he/she is notified of updates how updates impact their usage.
Use Case Cross References	
Includes	N/A
Extends	N/A

UC 7- 10

Use case ID 007 Use case Name: Fetch Timetable from Database	
Priority Medium	
Actors:	
Use Case Summary	The user's entered gesture to start following the timetable. User will be directed to the next scheduled slot
Pre-condition:	User has given the Input of the gesture

Normal Flow of Events	**Alternative Path**
2. The system will check within the database for the next room where is the class	User is notified that there is no class in the next slot.

UC 8- 11

Post Conditions	
Step#	**Description**

1.	User will start the navigation from the current location to the next class
Use Case Cross References	
Includes	None
Extends	None

UC 9- 12

Use case ID 009 Use case Name: Fetch Current Location	
Priority Medium	
Actors:	
Use Case Summary	User's current location is fetched from the map
Post Conditions	
Step#	**Description**
1.	Exact location of the user is fetched

Use Case Cross References	
Includes	None
Extends	None
Pre-condition:	User has given the Input of navigation
Normal Flow of Events	**Alternative Path**
2. The system will fetch the location of the user	

UC 10- 13

Use case ID 011 Use case Name: Custom Location	
Priority High	
Actors: User	
Use Case Summary	User wants to navigate to a custom location other than the scheduled timetable

Pre-condition:	User has given the Input of the Custom Location Gesture

Normal Flow of Events	**Alternative Path**
2. The system will start user's navigation from current location to the desired location	

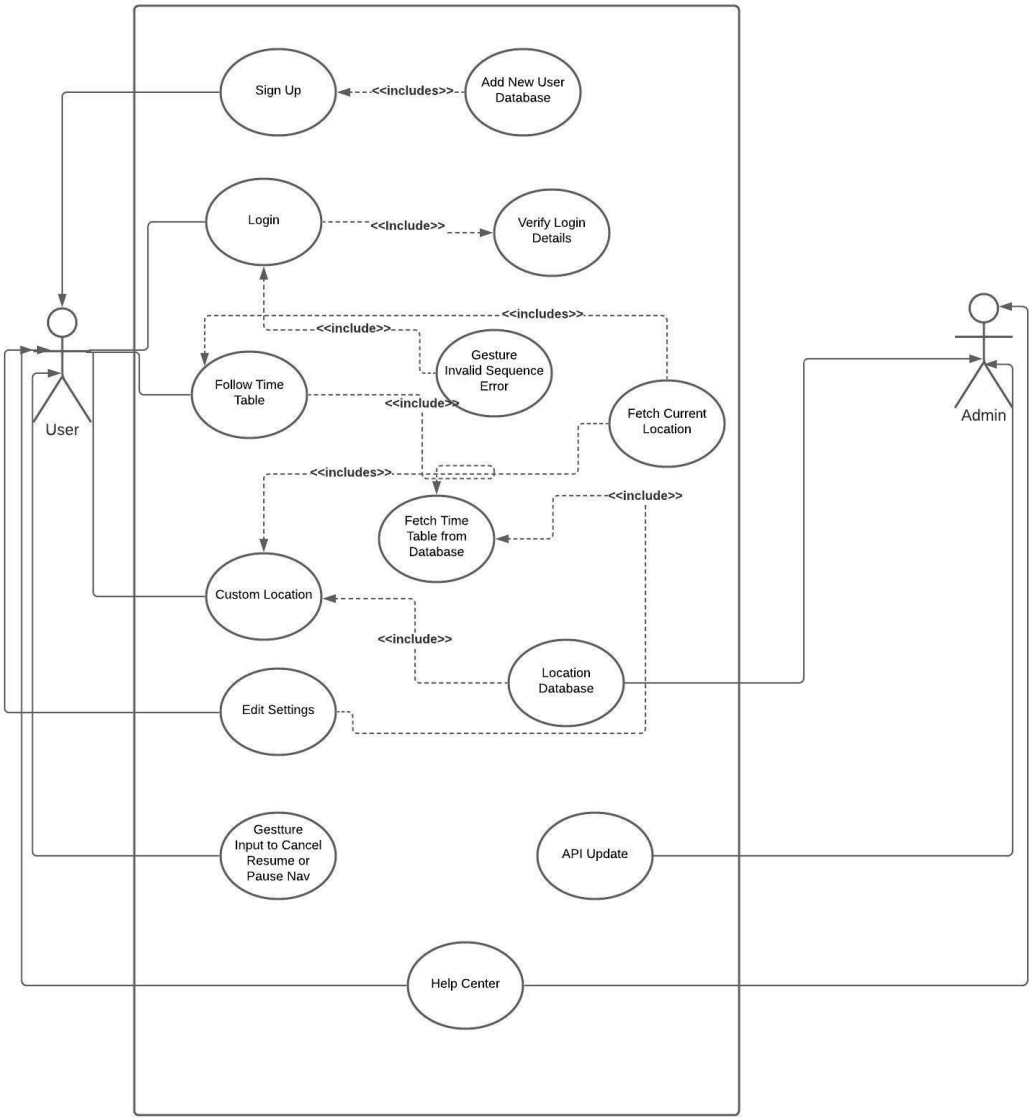
Post Conditions	
Step#	**Description**
1.	Navigation of the user starts from the current point to the location
Use Case Cross References	
Includes	Fetch Current Location Location Database
Extends	None

Use case ID 012 Use case Name: Location Database	
Priority Medium	
Actors:	
Use Case Summary	Exact location from the database is fetched of a particular location outside the study blocks
Pre-condition:	User has given the Input of the Custom Location Gesture
Normal Flow of Events	**Alternative Path**
2. Exact coordinated are fetched from the database of the desired location	

Post Conditions	
Step#	**Description**

1.	Exact Coordinates are fetched
Use Case Cross References	
Includes	None
Extends	None

2.5 Use case diagrams



1-

Use case diagram

2.6 Software development Life Cycle

<u>V shaped model</u>

For our project, we planned to go with V shaped model. This model was chosen for several reasons which are mentioned below. In this model, the process happens in a sequential manner. This model is also known as Verification and Validation model.

Reasons:

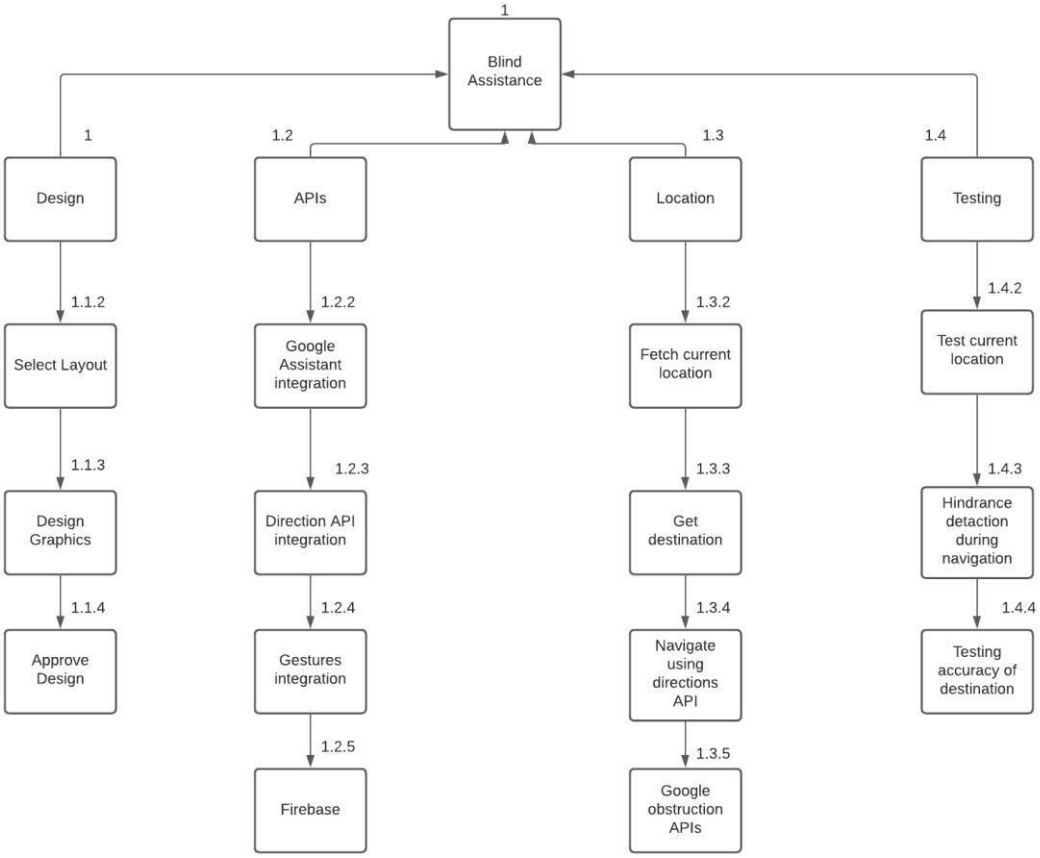
- The requirements for our app are well defined and known to us.
- Development tools and software are well known and decided.
- Each phase/step has its deliverables.
- Revisions with verifications of the app in each stage of development

3 System Design

The design figures at the high level are designed to show all the activities. Several entities have been recognized and in these diagrams the relationship between these entities is labelled. The various diagrams determined for this include:

1) Work breaks down structure (WBS)
2) Activity Diagram
3) Sequence Diagram
4) Class Diagram
5) System Level Use-Case

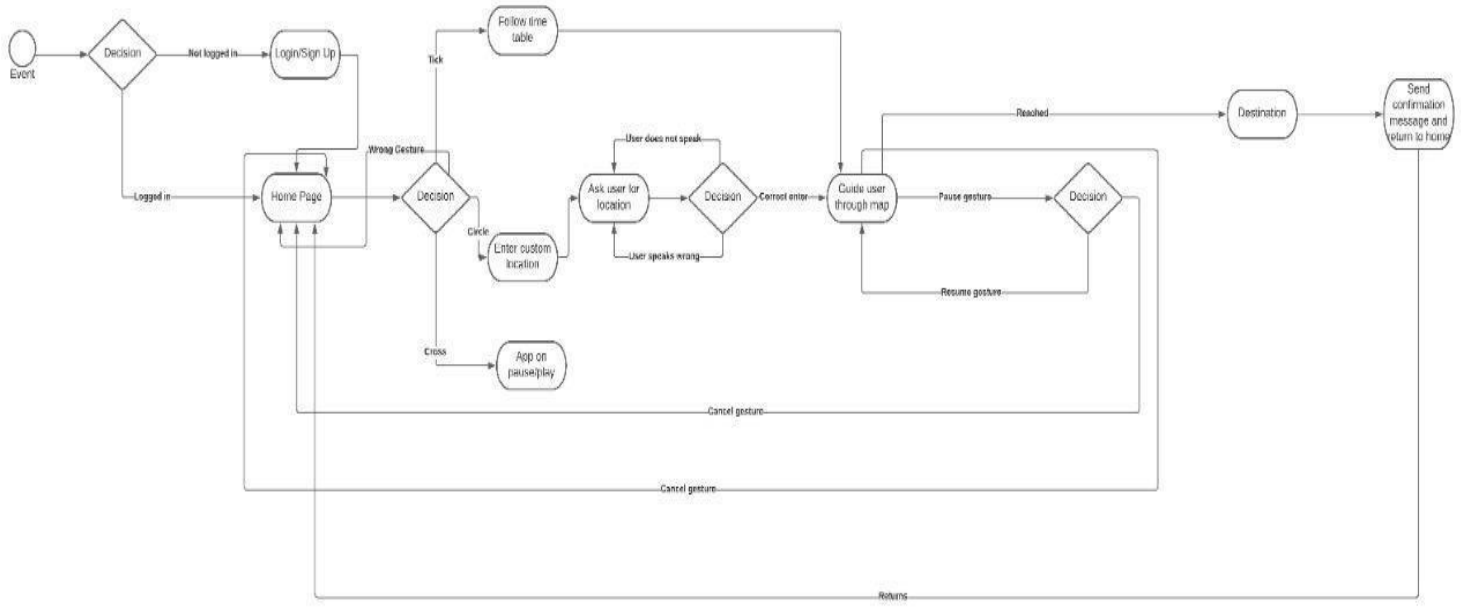
3.1 Work Breakdown Structure



2-Work break down structure

3.2 Activity Diagram

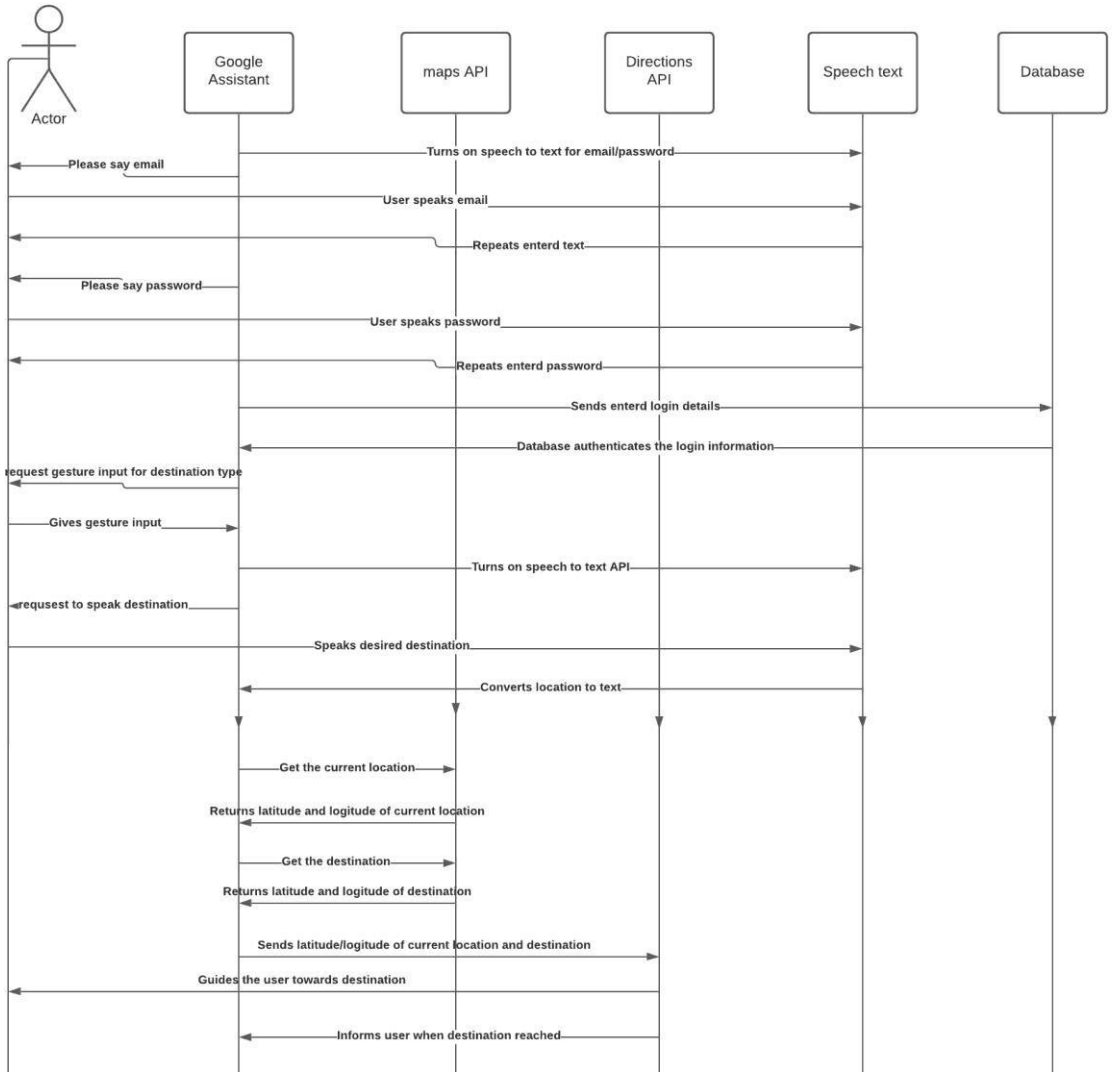
3.2.1 App activity diagram



***3**-App activity diagram*

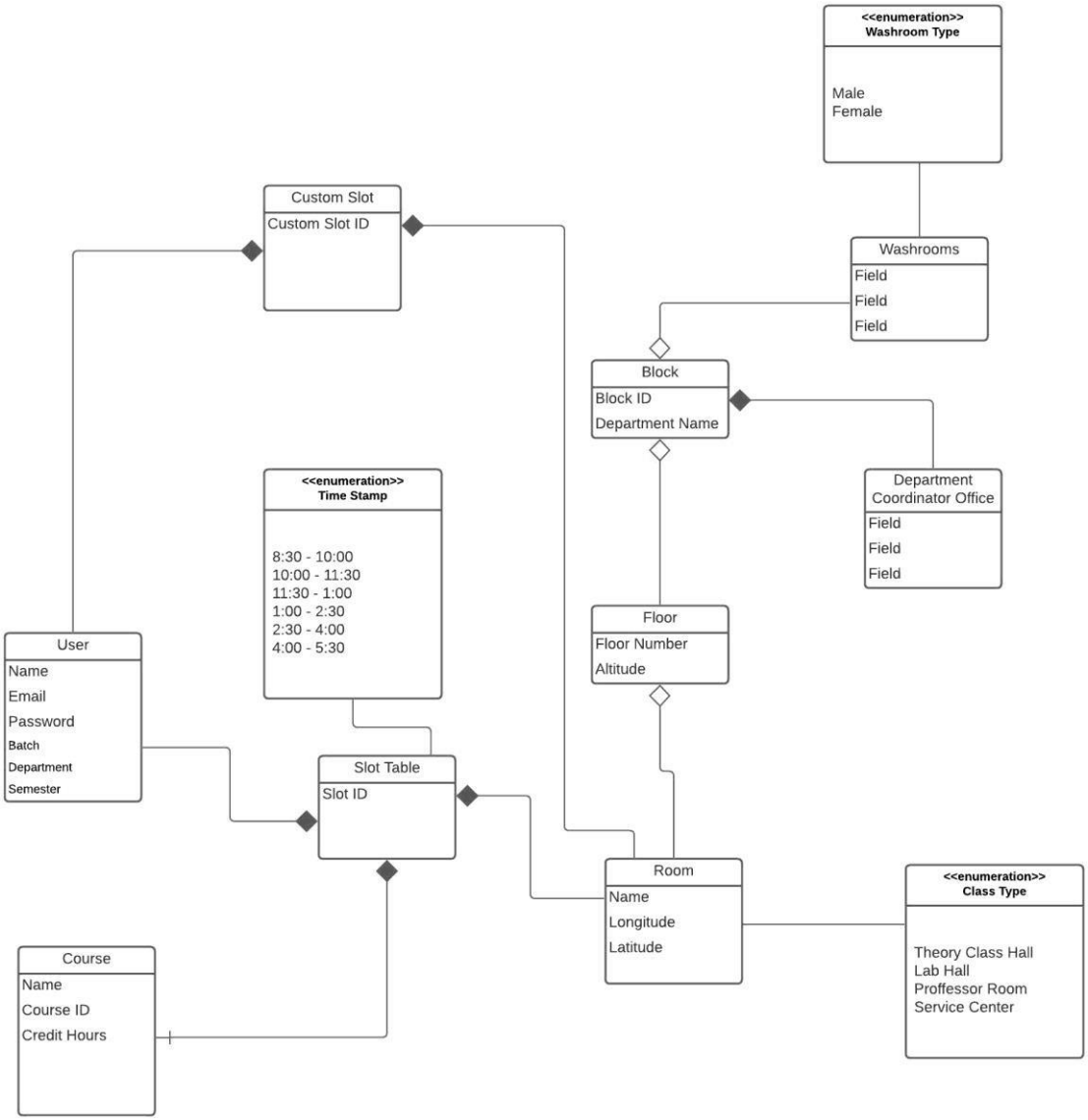
3.3 Sequence Diagram

3.3.1 User desired location Sequence diagram



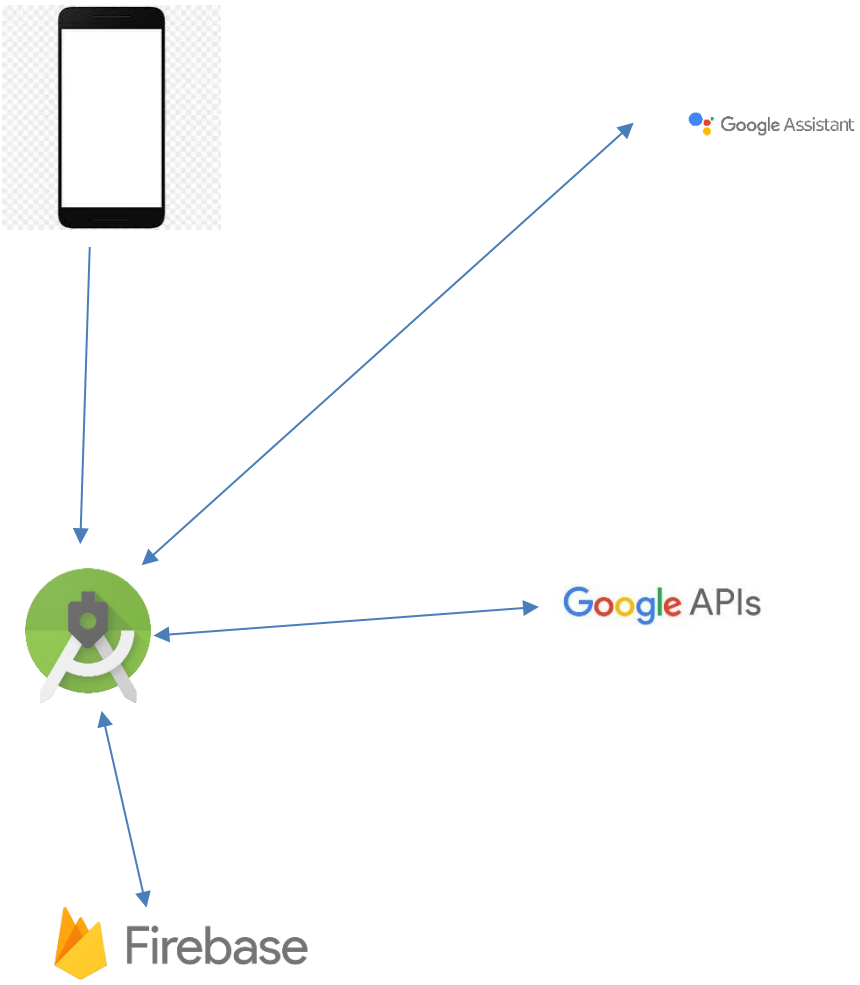
4-Sequence diagram

3.4 Class Diagram



5-Class diagram

3.5 System Architecture



6-System Architecture

4 System Testing

4.1 Test Cases

Test cases tables 15

Sr. No	Test case	Steps	Data	Expected result
TC 01	Open Application	–		Does not crash
TC 02	Login User	Enter User's Credentials	Authenticity from the database	Login Successful.
TC 03	Application Main Screen	Enter a gesture for. 1. Inside or outside N-block. 2. Go to desired location.	Get information from the database_	Speech and searching and editing successful
TC 06	User Enters a desired location	User inputs the desired location gesture then input's location through voice command	Desired location process invocation gesture and location	User is guided to the desired location

4.2 Unit / Integration / Acceptance Testing

4.2.1 Unit Test

Unit testing is a method of product development in which the smallest testable pieces of an implementation, called units, are tested for lawful operation individually and autonomously. To verify that there are no bugs and efficiency lags, each module is tested, built and will pass through number of tests to ensure that we end up with a quality product.

The modules to be tested are as following.

1) Application launch
2) Login/Signup
3) Entering Correct and incorrect gestures
4) Going for custom location

5) Checking if the anchors are working perfectly

6) Check voice activation.

4.2.2 Integration Test

In this phase, all the different programming modules are tested together as one. This phase comes after unit testing and before the Acceptance testing. In integration testing phase, we intend to test the full functionality to be executed.

The modules to be tested are as follows.

1. Each gesture worked correctly.
2. Path between two points is detected.
3. Voice command from both sides is working properly.
4. Anchors are loading properly
5. Database is updated at runtime of the application.

4.2.3 Acceptance Test

This test is conducted at the last just before deployment. In this test, app is used in a real time environment to check its usability and receive feedback from the people and fix bugs. We as the developers shall respond to this feedback carefully and fix the bugs immediately.

5 Conclusion

5.1 Problems Faced

Following were the problems faced while developing this app.

- During fetching of the location, different values for both longitude and latitude
- Implementation of custom gestures for user
- When anchors are loading, they can only be loaded from that direction from which they were placed. They won't load from any other direction.
  Solution:
  Every anchor has left, right value. If we are approaching the anchor from either direction, it loads that specific value.
- Anchors that have multiple anchors at its surroundings i.e., left, right back or front side, called a junction, didn't give proper path according to graph.
  Solution:
  Anchors that had "junction problems", we explicitly declared each anchor placed at their different directions i.e., front, back, left and right.
- Speech to text misreads in noisy environments.

5.2 Project Summary

A project that has focused solely on the humanity. Helping blind people navigate their way towards equal education to be able to work and leave their mark on the world. From the point of entry, to moving from any location to the other, this app shall be the best guide of the user. The app shall literally speak to the user and inform it about which turn to take and how to avoid the upcoming obstacle in the way. All shall be managed through this app.

5.3 Future Work

In the future, many newer features shall be added in this app. Users shall be able to use it in multiple locations and then ultimately throughout the city. Traffic condition shall be added on different routes.

References

[1] https://www.ariadnegps.eu/. n.d. Ariadne GPS. [online] Available at: <https://www.ariadnegps.eu/sample-page/> [Accessed 20 February 2021].

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[5] Evelity. 2021. *Home - Evelity*. [online] Available at: <https://evelity.com/en/> [Accessed 21 February 2021].

Appendix A

Turnitin Originality Report

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Blind Assistance By Omar Hashmi

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Class: FYP
Assignment: FA15-FYP report 2
Paper ID: 1141368407

1 Introduction 1.1 Introduction The Blind Assistance application is for Android only. Blind Assistance targets the 'Special people' as an audience, Blind or visually impaired people to be specific. Blind Assistance shall help the visually impaired people to either move from one desired point in the campus to another point or simply follow their specific class timetable and navigate towards their class at the specific time. Blind Assistance application mainly communicates back and forth with the user through voice; however, some inputs are given using gestures as well. As the Blind Assistance application is opened, the user shall be prompted to enter sign up details. This shall only be done once and saved for future use until the user wishes to use a different account altogether. Now the user would be at the 'Home Page'. Here the user shall have a choice to either go to a desired location or follow the timetable, he/she would have to draw the relevant gesture for desired option. Whatever option is chosen by the user, the user shall be taken to the desired location through 'speech' guidance using Google Maps and/or AR core API using anchors. 1.2 Goals and Objectives Following are the goals and objectives for our Blind Assistance Final Year Project. • • Benefit special people communities. Help each partially or completely blind person to be able to move through campus like any other person, without any kind of outside help. • • • Follow classes according to the timetable. Go to parking from working space and vice versa. Enable special people to be able to earn and support their families by getting equal education without any hindrances. • Enable blind person to move